What is ESTNeT?

ESTNeT is a **discrete-event simulator for space-terrestrial networks** with a fully implemented power model. Solar panel output and battery state are computed from orbital geometry and sun angle — making energy behavior physically grounded. The trade-off: it does not run real software, so networking behavior is less representative of actual satellite operation.

✅ Strengths

- Full power model: solar panels + battery
- Power tied to orbit geometry and sun angle
- Per-satellite specification support

⚠️ Limitations

- Does not run real routing software
- Less representative of real satellite behavior
- Discrete-event, not container-based

Equation 1: The Simple Model

The first equation models solar panel output as a function of sun angle. It is compact, requires only satellite and sun position, and directly answers the core question: *how is the battery charged based on panel orientation?*

The Formula

$$P = P_{max} \times \cos(\gamma)$$

Where γ is the angle between the sun vector and the solar panel normal. Output is maximum when the sun hits directly ($\gamma = 0$), decreases at oblique angles, and drops to zero during eclipse.

What It Needs

- Satellite position (already computed)
- Sun position (already computed)
- P_{max} — a single scalar per satellite

How to Extend

One extension point only: the Topology Configurator. No static config changes, no container modifications.

Equation 2: The Full Power Model

Expands Equation 1 with physical satellite specifications, enabling **per-satellite power modeling**. Each satellite can have its own panel area, cell efficiency, and system loss profile — matching real hardware diversity across a constellation.

$$P = A \times \eta_{sc} \times B_0 \times \eta_{sys} \times \cos(\gamma) \times (1 - \rho(\gamma))$$

A — Panel Area

Physical solar panel surface area in m².

Defined per satellite in static config.

η_{sc} — Cell Efficiency

Solar cell conversion efficiency, typically 25–30% for space-grade cells.

B_0 — Solar Constant

Fixed at ·1361 W/m² at 1 AU. No computation needed.

η_{sys} — System Efficiency

Accounts for wiring, regulator, and conversion losses across the power subsystem.

$\rho(\gamma)$ — Reflectivity Loss

Angle-dependent loss from panel surface reflectivity. Increases at grazing angles.

OpenSN can be extended in 3 Ways

Container Images

Defines what software runs inside each satellite. For v2, the container tracks real battery state and can perhaps shut down based on energy availability.

Topology Controller

The simulation brain: it decides satellite positions and inter-satellite links (ISL) each loop.

Static Config Files

Defines the simulation's initial shape. For v2, each satellite's panel area, efficiency, and system specs are declared here

Mapping Equations to Extension Points

Equation 1 is a minimal addition. Equation 2 uses the full extension — but each piece builds naturally on the last.

EQUATION 1 — V1

Simple Model Mapping

- **Input:** Satellite position + sun position
- **Computed in:** Topology Controller only
- **Config changes:** None
- **Container changes:** None

A small, self-contained addition. The Topology Controller already has everything it needs.

EQUATION 2 — V2

Full Model Mapping

- **Static Config:** Panel area, η_{sc}, η_{sys} per satellite
- **Topology Controller:** Computes power using each satellite's own specs
- **Container Images:** Track real battery state, respond to energy levels

Uses all three extension points together for full per-satellite energy modeling.